Advaith Moholkar

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Education

Indian Institute of Technology (IIT), Jodhpur — *Aug. 2023 – May 2027*
B.Tech. - Majors in Computer Science and Engineering (CSE) — **CGPA: 8.03/10**

Jayshree Periwal High School, Jaipur — *Apr. 2021 – May 2023*
Senior Secondary — **Percentage: 95.6/100**

Technical Skills

- **Programming & Software Engineering:** Python, C/C++, SQL (PostgreSQL, MySQL), JavaScript, HTML/CSS, Git
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, scikit-learn; CNNs, embeddings, transformer-based models
- **Data Analysis & Prototyping:** NumPy, Pandas, Matplotlib, Seaborn, Streamlit, Excel

Projects

LaneNet: Enhanced Curved Lane Detection — *GitHub*
(Mentor - Dr. Himanshu Kumar) — *Jan. 2025 - May 2025*

- *Developed **CLRLaneNet** by integrating a learned label-assignment network (**LaneNet**) into **CLRNet**, achieving a **+2.8% F1 improvement** on the **CULane benchmark** (100K+ images, 9 categories).*
- *Replaced heuristic cost-based matching with a **fully connected encoder–decoder**, improving generalization under real-world conditions such as curvature, shadows, and low-light scenarios.*
- *Conducted systematic **model evaluation and error analysis** and refined the architecture using **transfer learning** on 8,677 curved-lane samples.*
- **Tools & Technologies:** Python, PyTorch, CLRNet, ResNet, OpenCV, Albumentations, W&B

Smart Product Pricing — *GitHub*
(Amazon ML Challenge '25) — *Oct. 2025*

- *Built a **multimodal ML system** combining textual and visual features to predict e-commerce prices, achieving a final **SMAPE of 48.94** on a large-scale dataset and getting ranked in the **Top 300** out of 82K+ participants.*
- *Engineered modality-specific encoders using **EfficientNet-B2** and **ConvNeXt-Tiny** for images, and **Sentence-BERT**, TF-IDF, and rule-based features for text.*
- *Benchmarked multiple models and optimized performance via **LightGBM ensembling**, **weighted blending**, and **5-fold CV**.*
- **Tools & Technologies:** Python, PyTorch, LightGBM, Hugging Face Transformers, NumPy, Pandas

QuantSim: Quantitative Factor Investing Simulator — *GitHub*
(Self Project) — *May 2025 - Jun. 2025*

- *Developed a **multi-factor investment simulator** using **Random Forest models** and **FinBERT-based sentiment signals**, achieving **12.2% net returns** (Sharpe 0.68) on a 500-stock dataset (2010–2016).*
- *Designed a modular **ML pipeline** with data ingestion, model inference, and evaluation components, supporting **risk-adjusted allocation** via inverse-volatility weighting.*
- *Built an interactive **Streamlit prototype** to visualize factor behavior, portfolio allocation, and performance metrics for user validation.*
- **Tools & Technologies:** Python, Pandas, NumPy, scikit-learn, FinBERT, PostgreSQL, Streamlit

Financial Risk Assessment — *GitHub*
(Mentor - Dr. Avinash Sharma) — *Oct. 2024 - Nov. 2024*

- *Built a **credit risk scoring system** using **Random Forest, XGBoost, SVM**, achieving **98.8% accuracy**.*
- **Tools & Technologies: Python, scikit-learn, Flask, Streamlit, Pandas, NumPy, Matplotlib**.

Relevant Coursework

- Probability, Statistics and Stochastic Processes*(A)* ; Machine Learning*(A)* ; Linear Algebra and Calculus*(A-)* ; Design and Analysis of Experiments*(A-)* ; Data Structures and Algorithms*(A-)*

Certifications

- Amazon ML Summer School '25 — View Certificate
- Machine Learning Specialization - DeepLearning.AI — View Certificate